

Assignment #4 (Part I)

Course: ISA 345

Instructor: Dr. Maliha Alam

Points: 30

Submission instructions: this assignment is to be done individually. All your answers must be in a single document, *e.g.*, a *doc* or a *pdf* file. For database-modeling questions, you are allowed to take print screens of your models and insert the resulting figures into your document. Another option is to export diagrams to figures by clicking at “File -> Data Modeler -> Print Diagram -> To Image File” in Oracle SQL Developer, and then import these figures into a text editor such as MS Word. Data-modeling questions must be answered using the Data Modeler tool inside Oracle SQL Developer. Reverse engineering is allowed (*i.e.*, creating diagrams from DDL). Unreadable models will receive zero points. Submit the final document on Canvas before the due date.

Question 1: After observing the growth of social networks aimed at dog and cat owners (yes, they exist!!), you spot a tremendous business opportunity: *a social network for reptile owners*; after all, they seem to be more popular than dogs (<http://www.telegraph.co.uk/news/earth/3500882/Reptiles-now-more-popular-pets-than-dogs.html>). In this social network, a reptile can be friend of many other reptiles. However, there are reptiles with no friends whatsoever (anti-social reptiles ☺). That said, the above domain involves two business rules:

1. A reptile can be friend of many other reptiles
2. A reptile can be friend of nobody

The following questions concern the friendship relationship.

- a) Your first approach to model friendship is by using a single entity called REPTILE. For simplicity's sake, assume that there are only two attributes in that entity, namely *ReptileID* and *FriendID*. Try to model the friendship relationship using Data Modeler and one single entity. [7.5 points]

- b) Assuming that the attributes *ReptileID* and *FriendID* are of type VARCHAR, create the above table using SQL. After creating the table, try to insert the following data into your table: *TheLizardOfGEICO* is friend of *TheLizardOfRANGO*, (i.e., *ReptileID* = "*TheLizardOfGEICO*" and *FriendID* = "*TheLizardOfRANGO*"), and *TheLizardOfGEICO* is friend of *MasterViperFromKungFuPanda* (i.e., *ReptileID* = "*TheLizardOfGEICO*" and *FriendID* = "*MasterViperFromKungFuPanda*"). Which business rule is violated and why? [7.5 points]
- c) In your second attempt to model friendship, you decide to use two entities: REPTILE and FRIENDSHIP. For simplicity's sake, assume that there is only one attribute in REPTILE, namely *ReptileID*, and two attributes in FRIENDSHIP, namely *Reptile_1* and *Reptile_2*. All attributes are of type VARCHAR. Think about the table FRIENDSHIP as modeling some sort of contact list. For example, in the table below *TheLizardOfRANGO* and *MasterViperFromKungFuPanda* are in the contact list (are friends of) *TheLizardOfGEICO*, and *MasterViperFromKungFuPanda* is in the contact list (is friend of) *TheLizardOfRANGO*

FRIENDSHIP Table

Reptile_1	Reptile_2
<i>TheLizardOfGEICO</i>	<i>TheLizardOfRANGO</i>
<i>TheLizardOfGEICO</i>	<i>MasterViperFromKungFuPanda</i>
<i>TheLizardOfRANGO</i>	<i>MasterViperFromKungFuPanda</i>

Model the above setting using Data Modeler. [7.5 points]

- d) Show that your new model fixes the problem(s) you found in b). Specifically, create both the tables REPTILE and FRIENDSHIP in SQL. Next, insert three reptiles (*TheLizardOfGEICO*, *TheLizardOfRANGO*, and *MasterViperFromKungFuPanda*) into the REPTILES table, and the following friendship data into FRIENDSHIP: *TheLizardOfGEICO* is friend of *TheLizardOfRANGO*, and *TheLizardOfGEICO* is friend of *MasterViperFromKungFuPanda*. Explain why no business rule is violated with the new model. [7.5 points]